

Table of Bond Dissociation Energies—Continued

Bond	Reaction	Dissociation energy, D°				Method, remarks	Date	Ref.	
		0 K		298 K					
		kJ/mol	kcal/mol	kJ/mol	kcal/mol				
C-I	$\text{CH}_3\text{I} \rightarrow \text{CH}_3 + \text{I}$			221.7	53.0	A.P.; M.S.	1959	179	
		230	55			Previous data on $\text{HI} + \text{CH}_3\text{I} = \text{CH}_4 + \text{I}_2$	1961	103	
				220.1	52.6	E.I.	1962	6	
				230 ± 4	55 ± 1	Equilibrium	1963	65	
						$\text{HI} + \text{CH}_3\text{I} = \text{CH}_4 + \text{I}_2$			
		227.2	54.3	232.6	55.6	Thermochemical;	1967		
		226 ± 13	54 ± 3	232.2 ± 13	55.5 ± 3	based on 9			
	$\text{CNI} \rightarrow \text{CN} + \text{I}$	335	80	339	81	Thermochemical; based on 9	1967		
C-N	$\text{CN} \rightarrow \text{C} + \text{N}$			808	193	A.P.; M.S.	1961	127	
				728 ± 13	174 ± 3	Shock wave;	1961	120	
						$\text{Kr} + \text{C}_2\text{N}_2$			
						Analysis of various data	1962	6	
						Effusion; M.S.	1962	162	
			812 ± 21	194 ± 5	816	195	Thermochemical;	1967	
			723.4 ± 9.6	172.9 ± 2.3			based on 9		
			732	175	732	175			
			730.1 ± 21	174.5 ± 5	732 ± 21	175 ± 5			
		$\text{CH}_3\text{NH}_2 \rightarrow \text{CH}_3 + \text{NH}_2$			335	80	A.P.; M.S.	1959	128
			331	79	Analysis of data	1966	8		
			335	80	Thermochemical;	1967			
			331 ± 13	79 ± 3	based on 9				
	$\text{CH}_3\text{NO}_2 \rightarrow \text{CH}_3 + \text{NO}_2$			243	58	Analysis of data	1962	6	
238.9		57.1	246.9	59.0	Thermochemical;	1967			
			247 ± 13	59 ± 3	based on 9				
C-O	$\text{CO} \rightarrow \text{C} + \text{O}$	1070.23	255.79	1074.83	256.89	Analysis of data	1962	6	
		1071.77	256.16	1076.38	257.26	Thermochemical;	1967		
		1071.94	256.2 ± 0.1	1076.5 ± 0.4	257.3 ± 0.1	based on 9			
		± 0.4							
	$\text{CO}_2 \rightarrow \text{CO} + \text{O}$	526.14	125.75	532.20	127.20	Thermochemical	1962	6	
		526.14	125.75	532.16	127.19	Thermochemical;	1967		
		525.9 ± 0.4	125.7 ± 0.1	532.2 ± 0.4	127.2 ± 0.1	based on 9			
	$\text{CH}_3\text{OH} \rightarrow \text{CH}_3 + \text{OH}$	374.5	89.5					1956	132
				381.2	91.1	A.P.; M.S.	1959	179	
				~ 377	~ 90	Photodissociation	1962	6	
				381 ± 8	91 ± 2	Thermochemical	1966	8	
				378.7	90.5	Thermochemical;	1967		
				377 ± 13	90 ± 3	based on 9			
			370.7	88.6					
		370.3 ± 13	88.5 ± 3						
$\text{HCOOH} \rightarrow \text{CHO} + \text{OH}$			~ 377	~ 90	Photodissociation	1962	6		
			400.4	95.7	Thermochemical;	1967			
			402 ± 13	96 ± 3	based on 9				